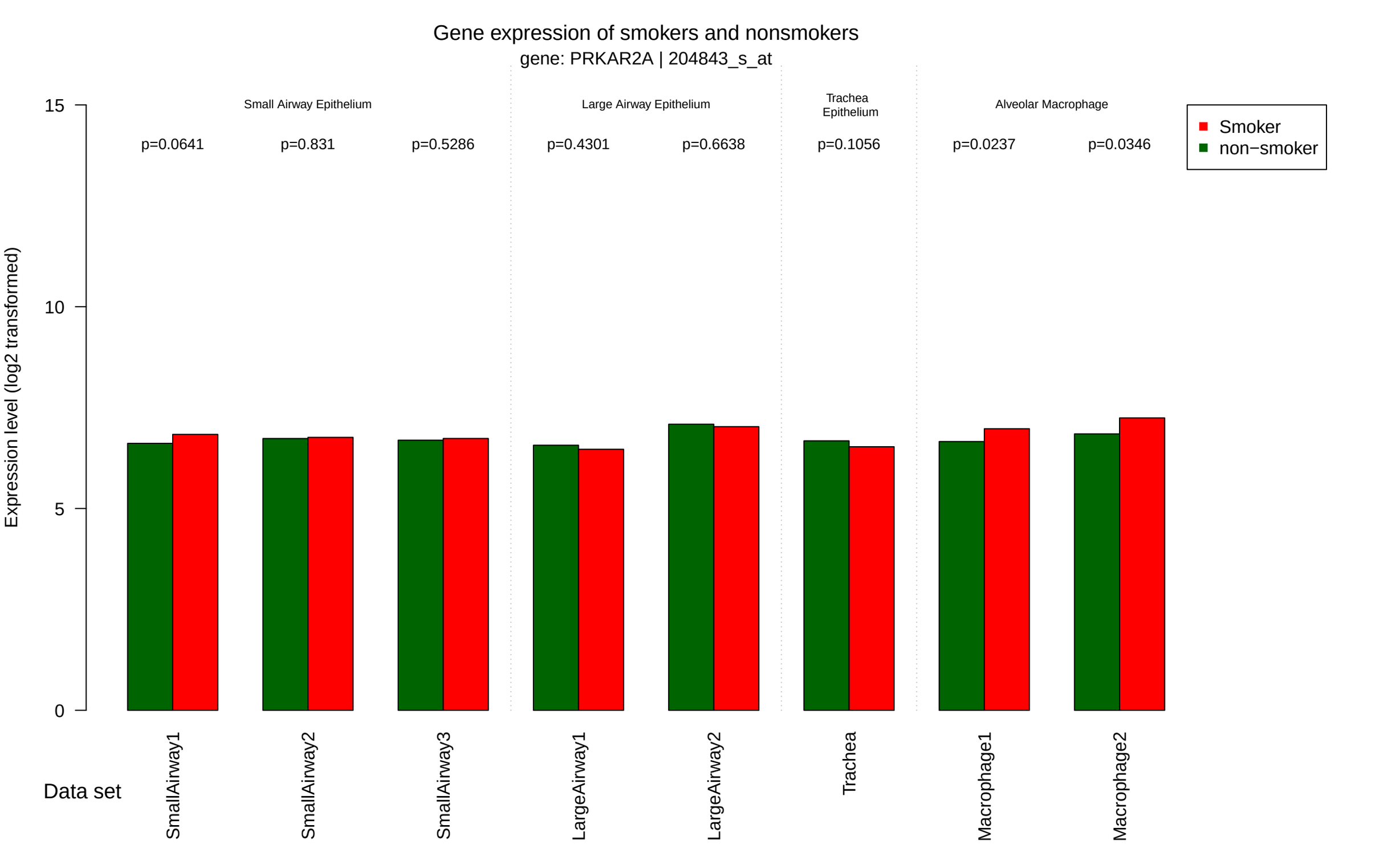
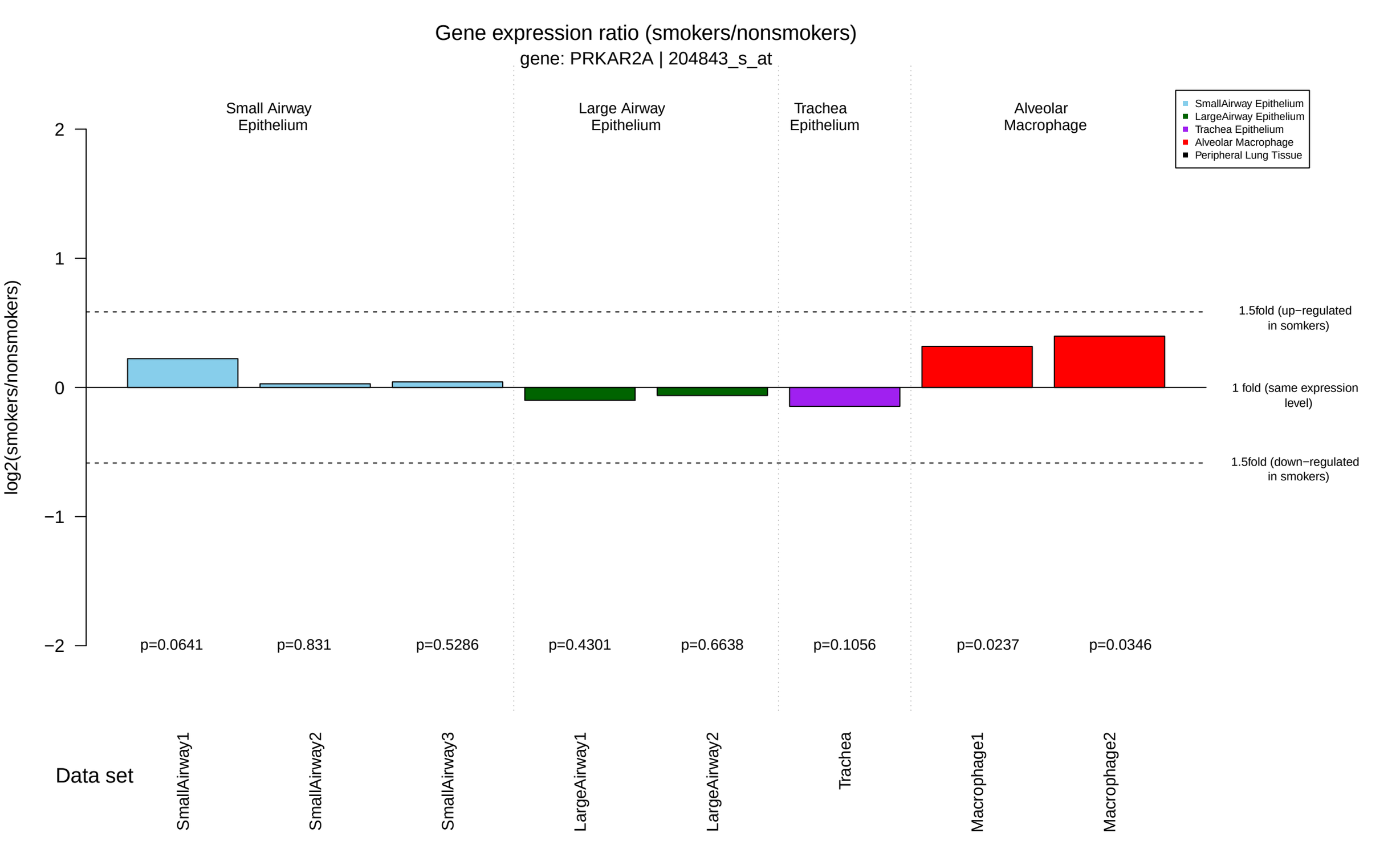
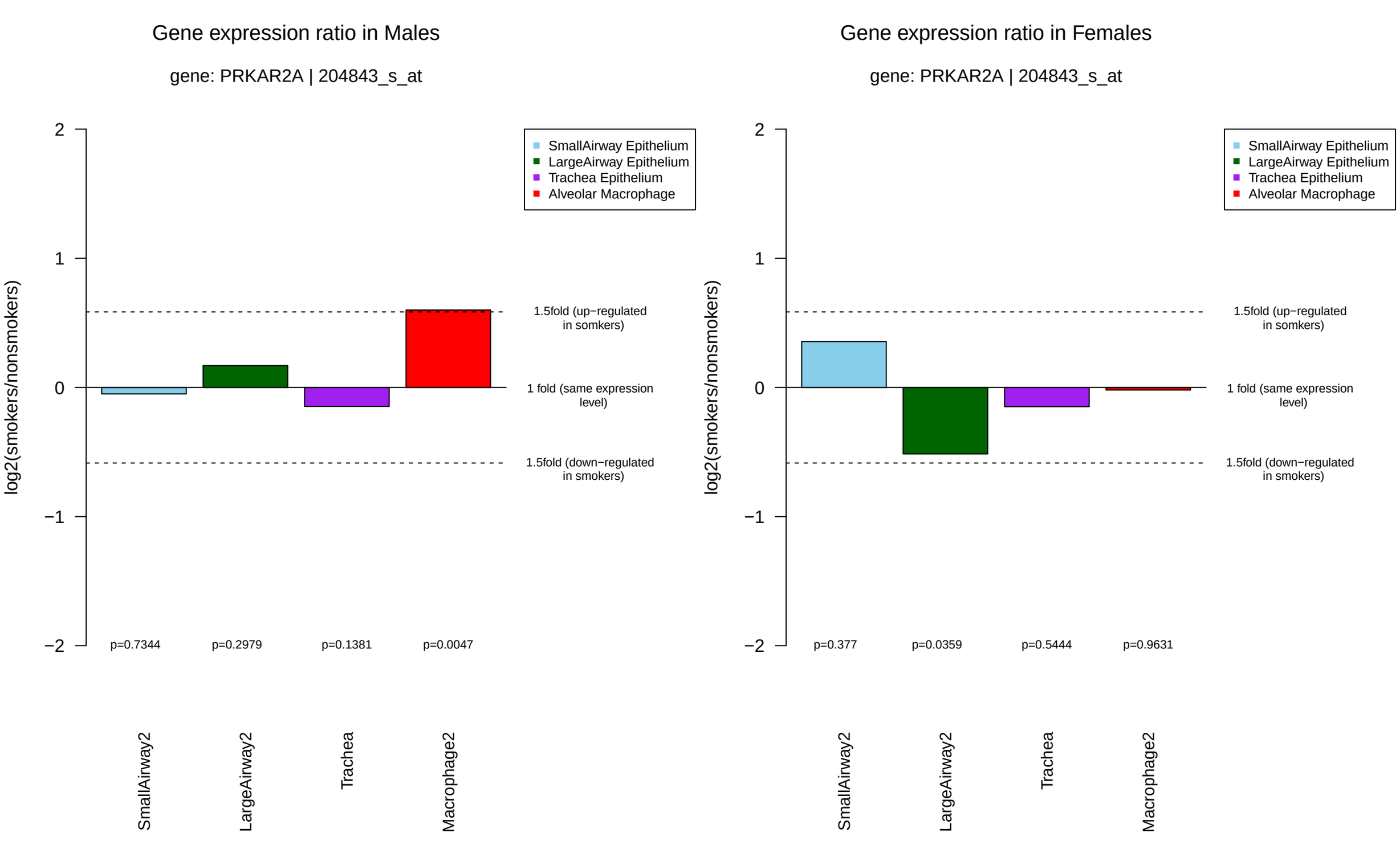
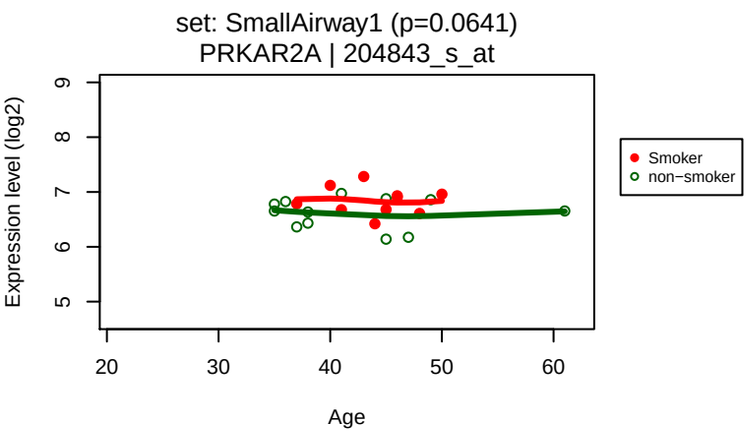
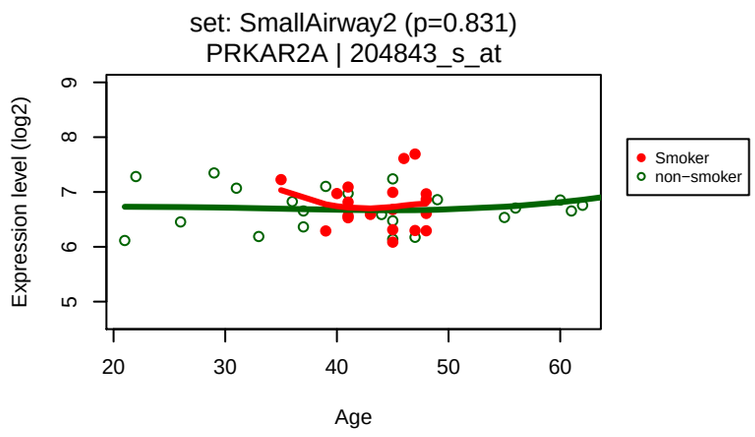
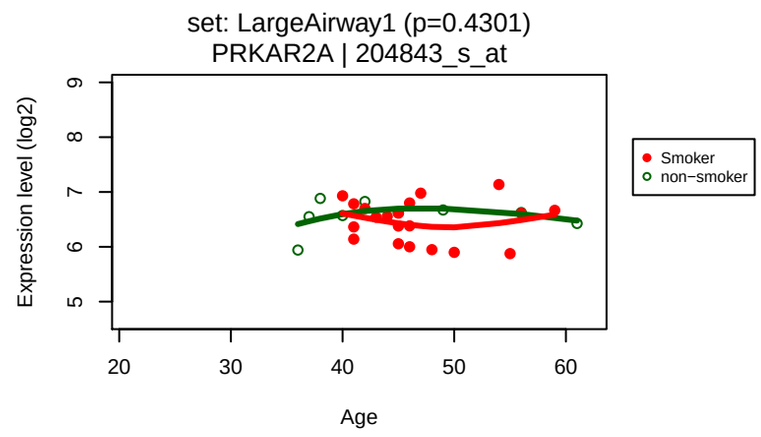

Gene expression of smokers and nonsmokers
gene: PRKAR2A | 204843_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0641
p=0.831
p=0.5286
p=0.4301
p=0.6638
p=0.1056
p=0.0237
p=0.0346
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

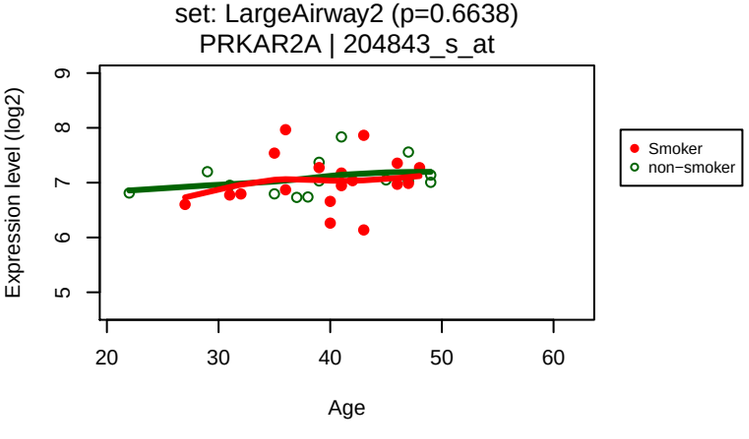

Gene expression ratio (smokers/nonsmokers)
gene: PRKAR2A | 204843_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0641
p=0.831
p=0.5286
p=0.4301
p=0.6638
p=0.1056
p=0.0237
p=0.0346
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

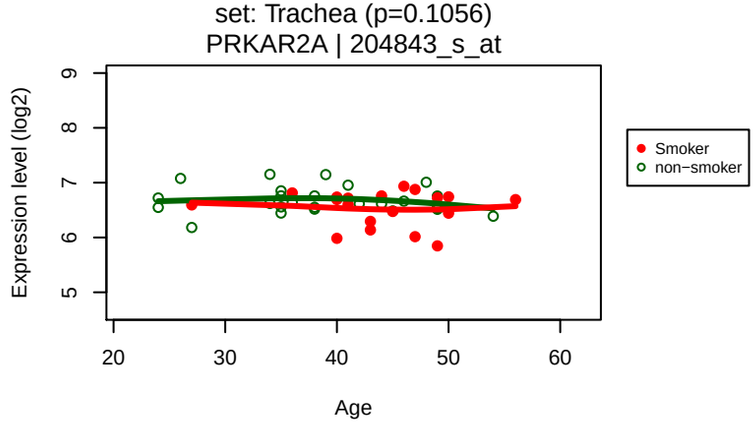

Gene expression ratio in Males
gene: PRKAR2A | 204843_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7344
p=0.2979
p=0.1381
p=0.0047
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAR2A | 204843_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.377
p=0.0359
p=0.5444
p=0.9631
SmallAirway2
LargeAirway2
Trachea
Macrophage2

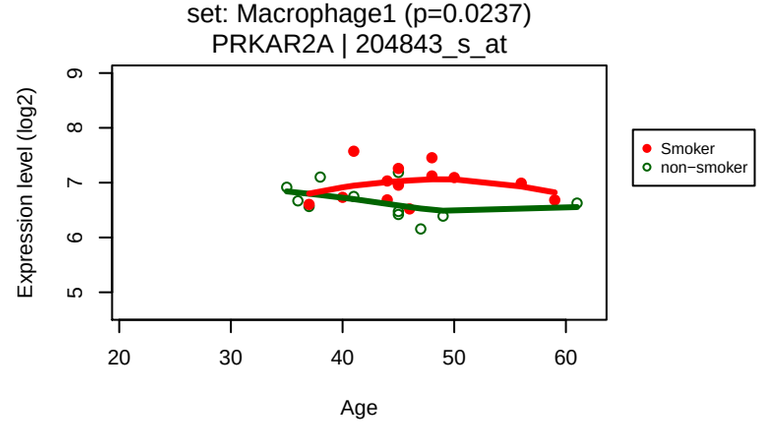

set: SmallAirway1 (p=0.0641)
PRKAR2A | 204843_s_at
Expression level (log2)
Age
Smoker
non−smoker

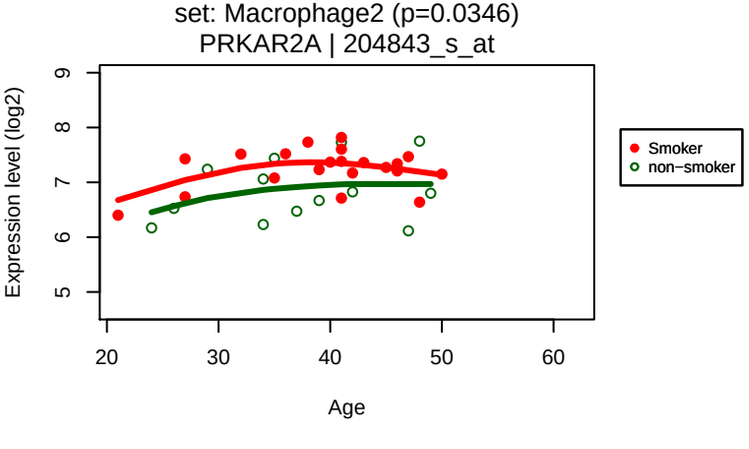

set: SmallAirway2 (p=0.831)
PRKAR2A | 204843_s_at
Expression level (log2)
Age
Smoker
non−smoker

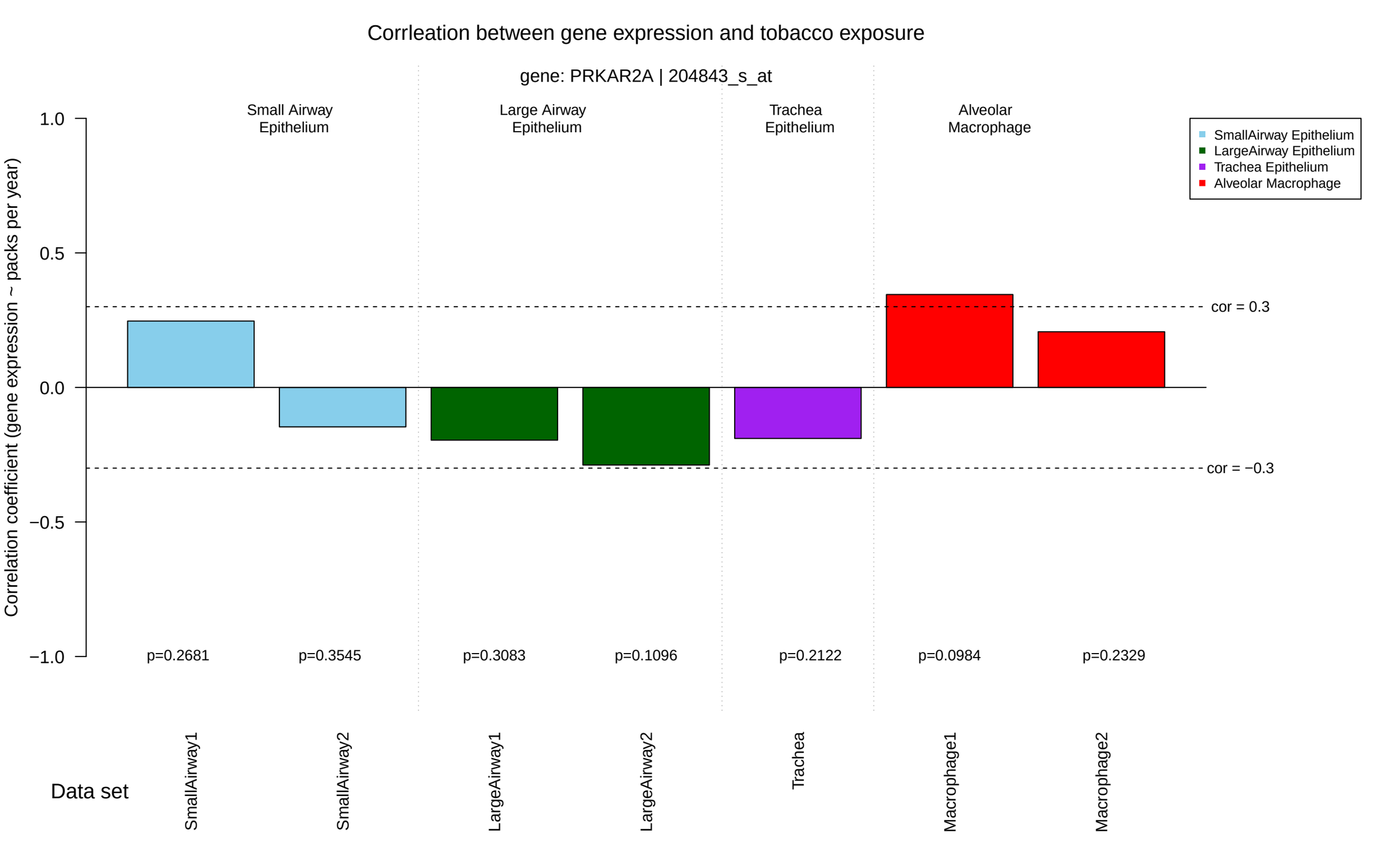

set: LargeAirway1 (p=0.4301)
PRKAR2A | 204843_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6638)
PRKAR2A | 204843_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1056)
PRKAR2A | 204843_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0237)
PRKAR2A | 204843_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0346)
PRKAR2A | 204843_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAR2A | 204843_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2681
p=0.3545
p=0.3083
p=0.1096
p=0.2122
p=0.0984
p=0.2329
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2